AWS BLUE-GREEN DEPLOYMENT

Project Documentation • Cloud & DevOps • Portfolio-ready summary

PROJECT GOAL
Demonstrate two application versions—Blue (Version 1) and Green (Version 2)—on AWS EC2, with separate environments to support controlled release and traffic switching.

1. Project Overview

This hands-on project demonstrates the basic Blue-Green Deployment approach using **AWS EC2** and **Nginx**. The Blue environment serves Version 1, while the Green environment serves Version 2. Separate versions make it easy to verify which environment is responding.

2. Technologies & Skills

- **AWS EC2** — virtual machines for hosting application environments
- **Linux / Ubuntu** — server administration and package management
- **Nginx** — web server used to serve the application pages
- **HTML** — simple version-specific deployment pages
- **AWS Security Groups** — network access rules for HTTP and SSH
- **Blue-Green Deployment** — parallel environments and release verification

3. Implementation Steps

- **Step 1 — Launch an EC2 instance:** Create an Ubuntu-based EC2 server and configure the required security group access.
- **Step 2 — Install Nginx:** Update package lists, install Nginx, and enable the service so the web server starts.
- **Step 3 — Deploy Blue Version 1:** Publish an HTML page identifying the Blue environment and verify it in a browser using the instance's public IP.
- **Step 4 — Deploy Green Version 2:** Publish a separate HTML page identifying the Green environment and verify that version in a browser.
- **Step 5 — Verify both versions:** Use the visible environment name and version number to confirm which deployment is serving each page.

4. Deployment Evidence — Blue Environment

The screenshot below shows the Blue environment serving Version 1 from the browser.

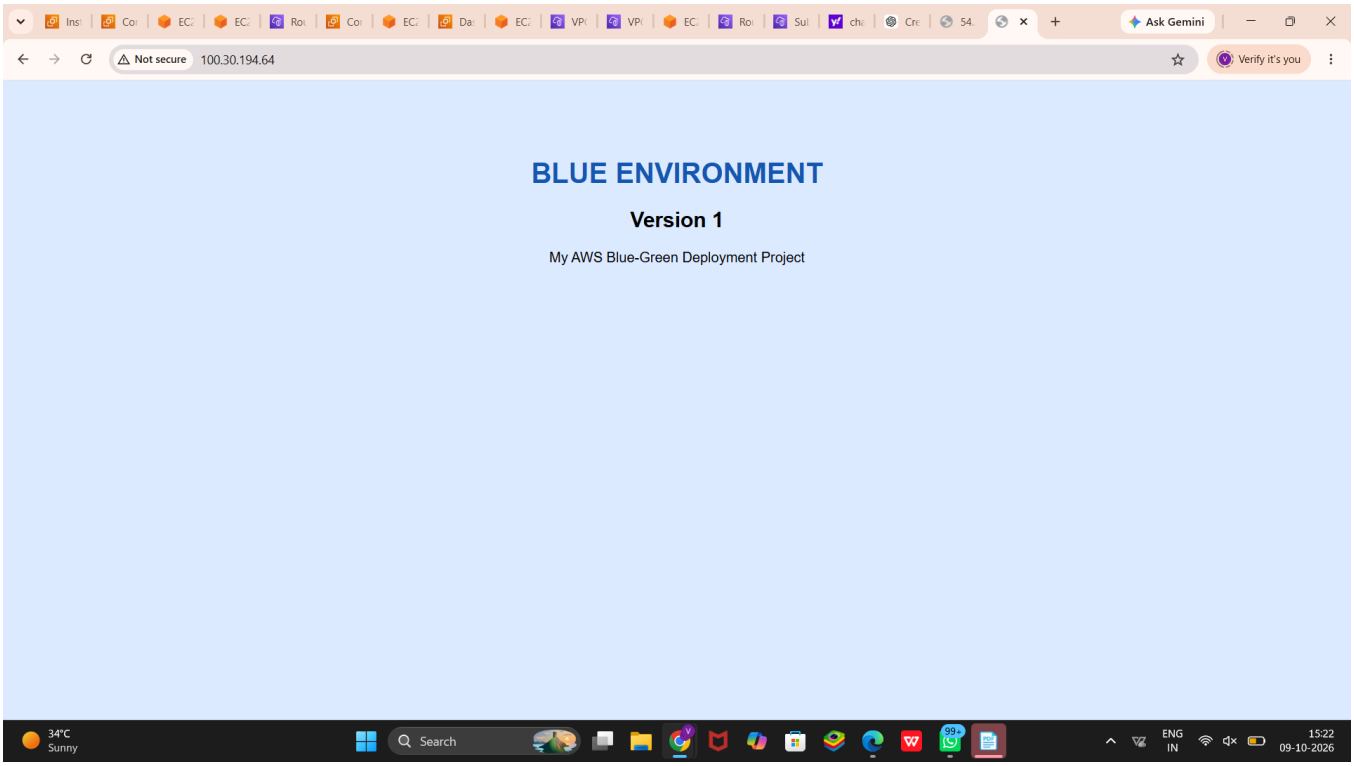


Figure 1. Blue Environment — Version 1 (browser verification).

5. Deployment Evidence — Green Environment

The screenshot below shows the Green environment serving Version 2 from the browser.

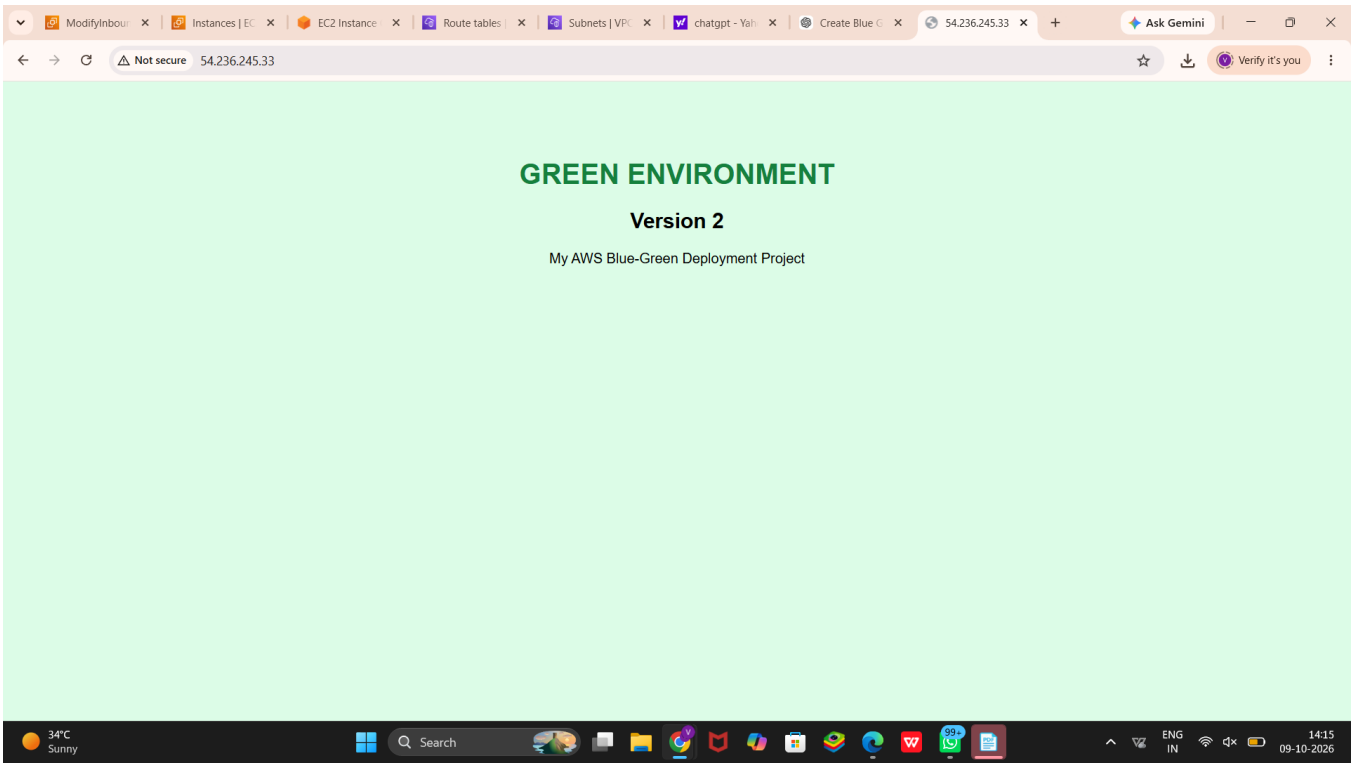


Figure 2. Green Environment — Version 2 (browser verification).

6. Outcome & Resume Description

Outcome: Successfully demonstrated two visually distinct application versions on AWS-hosted web servers and verified each through a browser. This provides a practical foundation for a full traffic-switching Blue-Green setup.

Resume-ready project entry

AWS Blue-Green Deployment | AWS EC2, Linux, Nginx, HTML

- Deployed versioned Blue (V1) and Green (V2) web pages on AWS EC2 using Nginx.
- Configured Linux server services and EC2 security-group rules to enable HTTP access.
- Verified each environment through browser-based testing, demonstrating parallel deployment environments and release validation.

Note: The screenshots verify the two pages are live. They do not by themselves prove that an Application Load Balancer or automated traffic switch was configured. Add those only after completing and testing them.